A Modular Machine Learning Framework for Automated Grievance Redressal System

A Project Report Submitted in partial fulfillment of the requirements
for the degree of Bachelor of Technology in Computer Science and Engineering

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Abstract

Public grievance redressal systems play a vital role in ensuring transparency, accountability, and responsiveness in governance. However, many existing systems still rely heavily on manual processing, which leads to delayed classification, inconsistent prioritization, and limited support for large-scale unstructured complaint data. This project addresses these limitations by proposing a modular grievance redressal analytics pipeline that transforms raw complaint text into structured and actionable information for administrative decision-making. The overall design follows a stage-wise workflow consisting of data acquisition, preprocessing, anomaly detection, domain classification, keyword extraction, severity risk scoring, resolution-time prediction, and sentiment analysis on feedback data.

The system first cleans and normalizes grievance records, then applies content validation and similarity-based duplicate detection to remove noisy, spam, abusive, irrelevant, or redundant inputs. After that, machine learning models are used to classify grievances into organizational domains, extract representative keywords, and estimate the urgency level of each complaint. In addition, a regression-based module predicts the expected resolution time, while a separate sentiment analysis component evaluates citizen feedback using weak supervision and supervised text classification.

The proposed framework demonstrates that grievance redressal can be treated as a structured natural language processing problem rather than only as a record-management task. By combining routing, prioritization, summarization, prediction, and feedback analysis, the system provides a more complete view of the grievance lifecycle and supports better operational planning. The notebook-based implementation with serialized model artifacts and intermediate outputs makes the workflow reproducible, extensible, and suitable for academic as well as practical use. Overall, the project presents an intelligent and modular approach for improving complaint handling, administrative decision support, and citizen service evaluation.

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Chapter 1

Introduction

1.1 Background and Context

Grievance redressal is one of the most important functions of any public service or citizen-support system. A grievance generally represents a complaint, request, concern, or expression of dissatisfaction raised by a citizen regarding a service, process, or administrative issue. In a governance environment, the manner in which grievances are received, analyzed, prioritized, and resolved has a direct impact on the quality of public service delivery, citizen satisfaction, and institutional credibility. A responsive grievance handling mechanism not only helps authorities identify recurring issues but also strengthens transparency, accountability, and trust between citizens and the administration.

In practice, however, grievance redressal systems continue to face several operational difficulties. Most systems receive a large number of complaints in free-text form, often written in informal language, with inconsistent terminology, spelling variations, incomplete information, emotional expressions, and repeated references to multiple issues within the same complaint. Such unstructured textual data is difficult to process manually at scale. Human-based sorting and routing of complaints may introduce delays, inconsistency, and subjective judgment, especially when a large number of grievances arrive every day. This makes it difficult for organizations to respond quickly and to ensure that serious complaints receive immediate attention.

Recent research has shown that natural language processing and machine learning can significantly improve the efficiency of grievance and complaint handling by automating the tasks of classification, prioritization, keyword extraction, and routing. Studies on AI-based grievance lodging and tracking, autonomous grievance redressal, complaint classification, civic complaint analysis, and complaint ticket categorization consistently highlight the limitations of manual systems and the advantages of intelligent automation in handling unstructured text data [1, 2, 3, 4, 5, 6]. These works collectively show that complaint text can be transformed into structured information that is useful for operational decision-making when appropriate text-mining and machine-learning techniques are applied.

The present project is built on the same core idea, but with a modular pipeline-based architecture that reflects the actual sequence of operations required in a grievance redressal workflow. The project is designed as a staged system in which raw grievance data undergoes preprocessing, anomaly detection, domain classification, keyword extraction, severity scoring, and resolution-time prediction. This sequential design is reflected in the project implementation and execution flow, where each stage contributes a distinct function to the overall system pipeline [7, 8].

1.2 Motivation of the Project

The motivation for this work arises from the need to build a practical and scalable grievance processing system that can support administrative action more effectively than manual workflows. Existing grievance portals often store complaint records, but the actual analysis of those records is limited. In many cases, grievances are merely collected and forwarded, without deeper processing of the complaint text to identify the concerned department, the urgency of the issue, or the expected time required for resolution. As a result, a large amount of potentially useful information remains hidden inside unstructured complaint records.

The literature indicates that automatic complaint analysis can address these challenges in multiple ways. Text classification models can identify the likely department or domain of a grievance, sentiment-aware or urgency-aware techniques can estimate its priority, and keyword extraction can summarize the main issue in a compact and interpretable form [9, 7, 8, 5]. At the same time, recent grievance-redressal research also shows that systems should be designed not only to classify complaints, but also to support operational intelligence such as trend analysis, routing support, and prioritization of high-impact issues [1, 2, 10]. This suggests that grievance redressal is not merely a data-entry problem; it is a structured decision-support problem.

Another important motivation behind this project is the need for interpretability and modularity. In many automated systems, the entire workflow is hidden inside a single model or a single application layer, making it difficult to analyze which component is responsible for a particular result. In contrast, the present project separates the pipeline into clear analytical stages. This makes the system easier to debug and easier to evaluate. It also makes the project more realistic, because grievance redressal in practice is naturally a multi-stage process rather than a single prediction task.

1.3 Problem Statement

The central problem addressed in this project is the development of an intelligent grievance redressal pipeline that can transform raw, unstructured complaint text into structured and actionable outputs. Specifically, the system should be able to identify noisy or anomalous records, preprocess and normalize complaint text, determine the appropriate complaint domain, extract important keywords, estimate severity or urgency, and predict the likely resolution time. These capabilities are needed so that grievances can be routed more efficiently and prioritized more appropriately.

The challenge is not only to classify a complaint, but also to understand its operational significance. For example, two complaints may belong to the same domain, but one may be far more urgent than the other. Similarly, a complaint may contain enough contextual information to predict that the issue will take longer to resolve. Therefore, the redressal process requires more than basic text categorization; it requires a pipeline of text analysis tasks that work together to produce a more useful administrative output.

In addition, the project keeps sentiment analysis on feedback data. It remains an important part of the overall system design because sentiment can offer additional insight into citizen dissatisfaction and can complement severity estimation. This makes sentiment analysis a structurally important component.

1.4 Objectives of the Project

The main objective of this project is to develop a modular AI-assisted grievance redressal system that can process complaint text through multiple analytical stages and provide meaningful outputs for grievance handling. The specific objectives of the work are as follows:

- to acquire and organize grievance-related data from the available datasets into a usable analytical format;

- to preprocess textual complaint records so that they become suitable for machine learning and natural language processing tasks;
- to identify and isolate anomalous, inconsistent, or low-quality records that may affect downstream predictions;
- to classify grievances into relevant domains or departments based on their textual content;
- to extract keywords and representative phrases that summarize the core issue in each complaint;
- to estimate the severity or urgency of a grievance using learned patterns from the data;
- to predict the expected time required for complaint resolution;
- to retain sentiment analysis on feedback data;
- to generate structured outputs that can assist in routing, prioritization, and reporting.

These objectives reflect both the practical needs of grievance management and the analytical directions demonstrated in the literature. Works on complaint classification and grievance redressal automation show that such tasks can be effectively handled using combinations of feature extraction, supervised learning, and rule-supported scoring approaches [3, 4, 5, 6]. The present project extends this idea by integrating these tasks into a single end-to-end pipeline.

1.5 Scope of the Work

The scope of the project is limited to text-based grievance analytics. The system operates on complaint data represented in textual form and focuses on extracting useful information from that text for downstream decision support. It does not attempt to replace the human administrative process entirely; rather, it aims to support it by improving the speed, consistency, and structure of complaint analysis.

Within this scope, the project includes data acquisition, preprocessing, anomaly handling, complaint domain classification, keyword extraction, severity estimation, and resolution-time prediction. The system is designed to be modular so that each stage can be independently developed and later combined into a unified pipeline. This modularity also makes the system suitable for experimentation and future enhancement.

The project does not focus on full-scale deployment, multilingual conversational interfaces, or visual evidence analysis in its current form. Those features may be considered as future extensions. Nevertheless, the current work provides a strong foundation for a practical grievance analytics engine and demonstrates how multiple NLP tasks can be chained together to support public-service workflows.

1.6 Organization of the Report

The remainder of this report is organized as follows.

- **Chapter 2** presents the literature review and discusses the major research contributions related to grievance redressal, complaint classification, prioritization, and text mining.
- **Chapter 3** describes the overall system architecture and the design of the proposed pipeline.
- **Chapter 4** explains the dataset, preprocessing strategy, and data preparation steps.
- **Chapter 5** details the proposed methodology module by module, including anomaly detection, domain classification, keyword extraction, severity scoring, resolution-time prediction and sentiment analysis.
- **Chapter 6** discusses the implementation details of the project codebase and pipeline execution.
- **Chapter 7** presents the results and evaluation of the developed system.
- **Chapter 8** concludes the report and outlines possible future enhancements.

Chapter 2

Literature Review

2.1 Introduction

The literature on grievance redressal and complaint analytics shows a clear evolution from manual complaint handling toward data-driven systems that apply natural language processing, machine learning, sentiment analysis, topic modeling, and, more recently, generative artificial intelligence. Across the reviewed studies, a common objective is visible: transform unstructured complaint text into structured operational information that can support faster routing, better prioritization, and improved service delivery. The papers considered for this project collectively address complaint classification, urgency estimation, domain identification, keyword extraction, and intelligent decision support, which are also the central design goals of the present system [1, 2, 3, 4, 5, 6].

For the present project, the literature was studied not as isolated prior work, but as a foundation for a modular pipeline. This is important because the implemented system does not solve a single NLP task in isolation; instead, it applies a sequence of analytical stages, namely data acquisition, preprocessing, anomaly detection, domain classification, keyword extraction, severity risk scoring, resolution-time prediction, and sentiment analysis component. The literature review is therefore organized around these functional stages so that it matches the structure of the codebase and the intended workflow of the project.

2.2 Evolution of Digital Grievance Redressal Systems

Early digital grievance systems primarily focused on collection and tracking of complaints rather than deeper analytical processing. Over time, research shifted from simple complaint logging toward systems capable of classification, redirection, prioritization, and status monitoring. The paper on *AI-Based Solution To Enable Ease of Grievance Lodging and Tracking* presents a centralized grievance platform that integrates natural language processing and machine learning to automate complaint submission, sorting, routing, and real-time monitoring across departments. It also introduces multilingual chatbot interaction, dashboards, automated alerts, escalation mechanisms, and predictive analytics, showing that grievance redressal can be strengthened through a combination of automation and decision support [1].

A similar direction appears in *Autonomous Grievance Redressal System for Government Services*, which emphasizes the limitations of conventional grievance systems such as manual prioritization, poor scalability, inaccurate domain categorization, and delayed response in cases involving visual evidence. That work extends the concept of grievance redressal by combining text and image data, thereby showing that grievance processing can benefit from multimodal inputs when complaints include supporting photographs or other visual evidence [2]. Although the current project is text-centric, this paper is relevant because it demonstrates the broader design direction of intelligent public grievance systems and identifies future opportunities for expansion.

The study *Smart Grievance Redressal System* also contributes to this progression by proposing a software application that identifies common issues, extracts keywords, groups similar complaints, and gives administrators a consolidated view of repeated grievance themes. Its use of text classification, keyword extraction, and keyword clustering demonstrates that grievance systems become more effective when they do not simply store complaints, but also summarize them analytically [8]. Taken together, these works establish that modern grievance redressal systems are increasingly expected to support structured interpretation of complaint text rather than merely archiving it.

2.3 Complaint Classification and Domain Identification

A major body of literature focuses on classifying complaints into domains or departments. This task is central to grievance routing because a complaint must first be assigned to the most appropriate service area before it can be resolved efficiently. In *Complaint Classification Model Using NLP*, the authors present a machine-learning based system that converts complaint text into numerical form using tokenization, stemming, lemmatization, count vectorization, and TF-IDF before applying classifiers for complaint categorization. The paper also discusses non-negative matrix factorization for grouping complaints into meaningful product or service clusters, showing that text classification can be combined with topic modeling to support routing and categorization [3].

The paper *Customer Support Ticket Categorization and Prioritization Using Natural Language Processing* is particularly relevant because it combines categorization and prioritization in a way that closely resembles the logic of the present project. It uses topic modeling through NMF to identify department-specific categories, while urgency and impact keywords are used to determine the priority level of tickets. This approach is valuable because it demonstrates that complaint classification is more effective when paired with a prioritization layer rather than treated as a standalone task [6].

Another closely related work is *Smart Civic Complaint Analyzer Using Natural Language Processing*, which develops a web-based system to categorize complaints into civic domains such as water, roads, electricity, and sanitation. The paper uses TF-IDF and Naive Bayes for classification and also extracts urgency and location information from complaint text. Its emphasis on practical deployment through a user interface and live data visualization shows that complaint classification is not only a modeling problem but also a systems problem involving usable interfaces and accessible analytics [4].

The work *NLP based Grievance Redressal System* similarly shows the value of using NLP and sentiment analysis to categorize railway-related social-media complaints and suggestions. It compares algorithms such as Naive Bayes, Decision Tree, Random Forest, and SVM, and demonstrates that unstructured grievance text can be classified effectively when suitable preprocessing and feature extraction are used [7]. These studies collectively support the present project's domain-classification module and justify the use of a staged text-processing pipeline.

2.4 Prioritization, Severity Estimation, and Sentiment Analysis

A second major theme in the literature is grievance prioritization. In public-service systems, classification alone is insufficient because not all complaints have equal urgency. Some grievances indicate immediate harm or service breakdown, whereas others are informational or low-priority in nature. The literature therefore frequently combines classification with severity estimation or sentiment analysis.

The paper *Civil Complaints Management System by using Machine Learning Techniques* proposes a grievance-handling model that applies sentiment analysis to citizen complaints in order to prioritize issues based on polarity and urgency. It uses machine-learning methods to categorize complaints into departments and emphasizes that severe complaints should be handled faster. This work is important because it connects emotional or sentiment-related information with administrative priority assignment [9].

The survey *Survey on Civil Complaints Management System by Using Machine Learning Techniques* reinforces this viewpoint by showing that sentiment analysis and clustering can help rank grievances according to urgency. It also reviews the use of classic machine-learning methods such as Naive Bayes and SVM for sentiment-driven prioritization, making it clear that prioritization is often derived from textual evidence rather than manual judgment alone [8]

Chapter 3

System Overview and Architecture

3.1 Introduction

The proposed grievance redressal system is designed as a modular, stage-wise analytical pipeline that transforms raw complaint text into structured outputs that can be used for administrative decision-making. Rather than treating grievance processing as a single classification problem, the system decomposes it into a sequence of dependent tasks. This design is consistent with the practical workflow of a grievance redressal environment, where incoming records must first be prepared, then checked for quality, categorized, summarized, prioritized, and finally assigned an estimated resolution time. The implementation of the project follows this principle explicitly through a chained execution structure, where each notebook produces the input required by the next stage.

This architecture is appropriate because it demonstrates both conceptual clarity and implementation maturity. Each module is functionally independent, yet the output of one module becomes the input to the next, thereby forming an end-to-end pipeline. Such a design makes the system easy to extend, debug, and evaluate.

3.2 Design Principles

The architecture of the system is guided by the following design principles:

1. **Modularity:** Each analytical task is implemented as a separate module, allowing isolated development and testing.
2. **Sequential dependency:** The output of each stage is preserved as an intermediate dataset and used in downstream stages.

3. **Interpretability:** Classical machine learning methods are preferred where possible, making the predictions more explainable.
4. **Scalability:** The use of notebook-based processing and serialized models enables the pipeline to be extended with minimal changes.
5. **Practical relevance:** The system is aligned with grievance redressal requirements such as routing, prioritization, severity assessment, and time prediction.

3.3 Overall Architecture

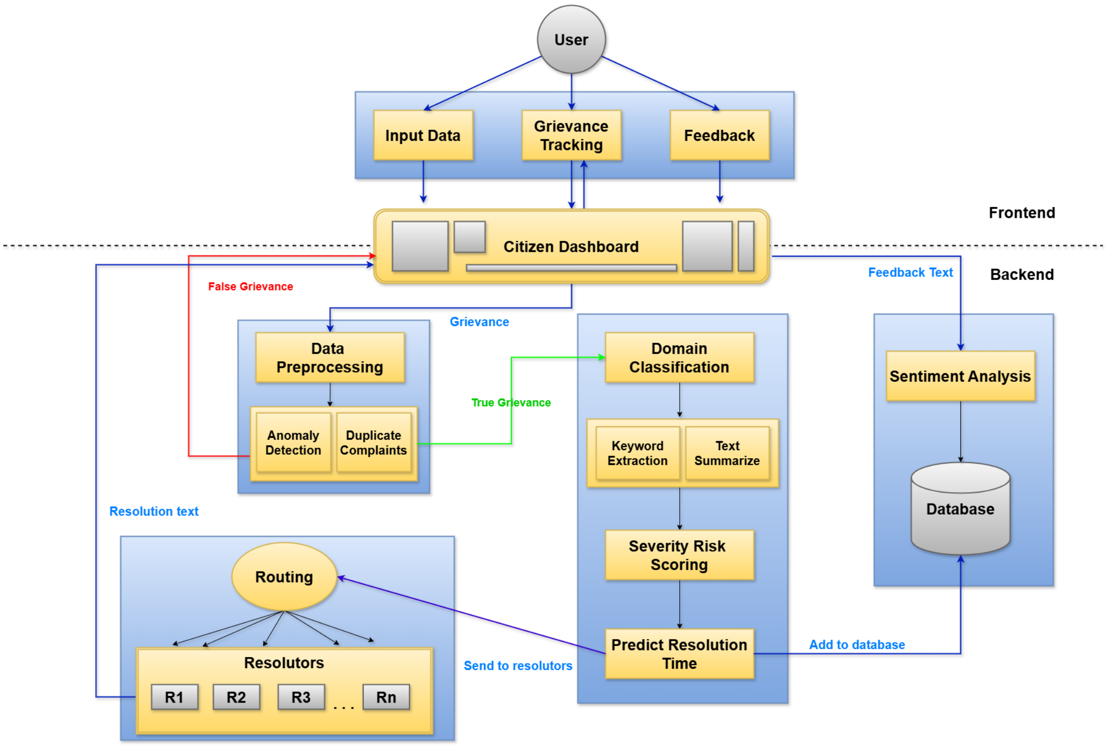

Figure 3.1: Overall architecture of the proposed grievance redressal system

3.4 Functional Workflow

The complete workflow of the system can be described as follows:

1. **Data acquisition:** Grievance records are collected from the available datasets and loaded into the system.
2. **Preprocessing:** The raw complaint text is cleaned and normalized to create a machine-readable representation.

3. **Anomaly detection:** invalid, spam, abusive, irrelevant, and duplicate entries are filtered using content validation, blocking, and TF-IDF cosine similarity.
4. **Domain classification:** The complaint is mapped to the most relevant organizational domain using a trained text classifier.
5. **Keyword extraction:** Salient terms and phrases are extracted to summarize the core issue.
6. **Severity risk scoring:** The urgency of the complaint is estimated using textual and contextual features.
7. **Resolution-time prediction:** The likely duration required to resolve the complaint is predicted.

3.5 Module-Level Description

3.5.1 Data Acquisition and Input Layer

The system uses grievance datasets stored in CSV form. These files contain complaint text and metadata such as receipt date, state, sex, and other administrative fields. The data layer is organized so that each stage of the pipeline reads an input file and writes a corresponding output file. This creates a reproducible chain of intermediate datasets.

3.5.2 Preprocessing Layer

The preprocessing layer performs text normalization and basic quality control. It prepares the complaint text for downstream tasks by removing noise such as HTML tags, URLs, punctuation, and stopwords, while also converting all text to lower case. A more rigorous filtering step is applied during training data preparation to retain only relevant grievance records.

3.5.3 Anomaly Detection Layer

The anomaly detection layer filters invalid or redundant grievance records before they are forwarded to downstream modules. It first applies a content validation step to reject spam, abusive, irrelevant, or too-short inputs, and then performs similarity-based duplicate detection using a time-aware blocking strategy and TF-IDF cosine similarity.

3.5.4 Domain Classification Layer

This layer predicts the likely complaint domain or department using a supervised text classifier trained on grievance text. The model is serialized and reused during the inference pipeline.

3.5.5 Keyword Extraction Layer

This layer extracts representative keywords and phrases using multiple extractive text-mining methods. It helps compress long grievances into compact semantic summaries.

3.5.6 Severity and Time Estimation Layer

The severity module predicts urgency class labels, and the time prediction module estimates how many days may be required to resolve the grievance. These outputs are especially useful for prioritization and administrative planning.

3.5.7 Sentiment Analysis Layer

The sentiment analysis module processes citizen feedback associated with grievances and classifies it into sentiment categories such as `satisfied`, `neutral`, and `unsatisfied`. Since feedback data is often unstructured, the module uses weak supervision to generate initial labels and then trains a text classifier on cleaned feedback data.

3.6 Technology Stack

The implementation uses the following technologies:

- **Python** as the primary programming language;
- **Pandas** and **NumPy** for data handling;
- **scikit-learn** for preprocessing, classification, and evaluation;
- **XGBoost** for regression and classification experiments in the time-prediction module;
- **YAKE**, **spaCy**, **TextRank**, and **LexRank** for keyword extraction and summarization;
- **Pickle** for model serialization;
- **Jupyter Notebook** for modular development and experimentation.

Chapter 4

Data Acquisition, Dataset Description and Preprocessing

4.1 Introduction

The quality of a grievance redressal system depends heavily on the quality of its input data. Complaint records are often noisy, incomplete, and inconsistent because they are written by different users in different styles, levels of formality, and levels of detail. For this reason, the project performs a dedicated preprocessing stage before any learning or inference module is executed. The preprocessing stage is not merely cosmetic; it is a necessary step to ensure that text-based machine learning models receive a normalized and meaningful representation of each grievance.

This chapter presents the data foundation of the proposed grievance processing system. Since the subsequent modules depend heavily on the quality and consistency of the input text, the data acquisition and preprocessing stages are treated as a core part of the methodology rather than a preliminary housekeeping step. The notebook-based workflow used in this project first converts the raw grievance records into a tabular format, then inspects the structure of the data, and finally applies a sequence of cleaning, filtering, and normalization steps to obtain a reliable dataset for downstream analysis.

4.2 Data Acquisition and Format Conversion

The analysis utilizes real-world grievance data provided in a semi-structured format. The primary dataset consists of a JSON file:

- `No_PII_Grievance_Data.json`: Contains the core metadata for individual grievances, including registration numbers, submission dates, and category codes. **Dataset Link:** Grievance Dataset

The raw grievance data is initially available in JSON form, including records that follow a MongoDB-style extended JSON representation. To make the data suitable for tabular analysis, the records are first normalized and then converted into a structured comma-separated format.

4.3 Attribute Description

The grievance dataset comprises several critical attributes required for analysis. The key fields identified during the initial inspection are:

Attribute	**Description**
`registration_no`	Unique identifier for the grievance
`org_code`	Code of the organization receiving the complaint
`recvd_date`	Timestamp when the grievance was submitted
`closing_date`	Timestamp when the grievance was resolved
`state`	State code of the complainant
`sex`	Gender of the complainant
`grievance_text`	Description of the grievance submitted by the citizen
`remarks_text`	Remarks text provided by the organisation

Table 4.1: Key Attributes in the Grievance Dataset

4.4 Initial Data Inspection

After loading the structured grievance data, the notebook performs standard inspection tasks to understand the size and composition of the dataset. These include checking the number of rows and columns, reviewing data types, examining sample records, and setting the unique identifier as the index. The columns are then reordered to produce a cleaner and more interpretable working table. This type of inspection is essential because grievance data often contains a mixture of text, dates, categorical labels, and incomplete metadata.

4.5 Exploratory Data Analysis

The acquisition notebook includes several exploratory visualizations that provide a first look at the distributional behavior of the dataset. These visual checks help identify skewed class distributions, geographic concentration, and temporal irregularities before preprocessing begins.

Table 4.2: Exploratory analysis views used in the acquisition notebook

Variable / View	Purpose	Analytical Value
State-wise grievance count	Measures geographic distribution of complaints	Identifies states with higher complaint volume
Gender distribution	Compares complaint counts by sex	Reveals demographic representation in the data
Organization code frequency	Examines frequency of complaint domains / departments	Indicates class imbalance and dominant categories
Daily grievance trend	Tracks complaints received over time	Helps understand temporal variation in complaint inflow
Closing-time distribution	Studies the duration between receipt and closure	Supports resolution-time analysis and anomaly detection

The state-wise distribution plot is especially useful because it indicates how grievance volume is distributed across different regions. Similarly, the gender distribution visualization provides a demographic snapshot of the complainant records available in the dataset. The organization-code frequency table is important because it reveals whether the dataset is balanced across departments or whether some categories dominate the corpus. The daily complaint trend and closing-time histogram are particularly relevant for later prediction tasks because they capture the temporal dynamics of grievance handling.

4.6 Data Cleaning and Quality Assurance

After the acquisition stage, the preprocessing notebook performs a stricter filtering pass to improve the reliability of the textual grievance records. Since raw complaint data often contains short entries, mixed-language records, incomplete entries, and low-information text, the notebook applies a set of explicit quality conditions before creating the final working dataset:

- the grievance text must be composed of ASCII characters, indicating English-language compatibility;
- the grievance text must be longer than 100 characters;
- records with `org_code` equal to "pending" are excluded.

This filtering step improves dataset consistency and removes records that are too short, ambiguous, or unsuitable for supervised processing.

4.6.1 Handling of Missing Values

The preprocessing notebook handles missing data in a selective manner. Missing remarks are replaced with a default placeholder, while critical fields such as closing date, state, organization code, and grievance text are not imputed and instead cause the affected row to be removed. This is a reasonable design choice because auxiliary remarks can be safely replaced, but essential fields should remain trustworthy for downstream reasoning.

4.6.2 Type Standardization

Before further processing, the grievance text is explicitly cast to string type so that text operations such as lowercasing, regex replacement, and token-based filtering can be applied safely. This type standardization reduces the chance of runtime errors and makes the text processing pipeline more stable.

4.7 Text Normalization

After filtering, the grievance text is copied into a separate processed column. The following transformations are then applied sequentially:

4.7.1 Case Normalization

All text is converted to lower case so that lexical variations such as "Water", "water", and "WATER" are treated identically by the model.

4.7.2 HTML Tag Removal

Any embedded HTML tags are removed using a regular-expression based cleaner. This ensures that formatting artifacts do not contaminate the linguistic content.

4.7.3 URL Removal

Links and web addresses are stripped from the text because they rarely contribute meaningfully to the complaint semantics and may introduce unwanted token noise.

4.7.4 Punctuation Removal

Punctuation symbols are replaced by spaces. This is particularly useful for complaint text, where punctuation may appear inconsistently and does not usually carry domain-specific meaning.

4.7.5 Stopword Removal

Common English stopwords are removed to reduce lexical noise. This improves the signal-to-noise ratio for models such as TF-IDF and SVM.

The cleaning pipeline is deliberately simple and interpretable. It does not rely on heavy linguistic transformations because the grievance corpus is expected to contain a large amount of practical, domain-specific language where a compact and robust preprocessing strategy is often more effective than an overly complex one. The cleaned text is stored separately from the original complaint text so that both raw and processed versions remain available for inspection.

4.8 Deduplication and Record Consolidation

After cleaning, the notebook applies deduplication based on the grievance text field. If multiple rows contain the same complaint text, only the first occurrence is retained and the rest are removed. This step is important for two reasons. First, it prevents redundant records from biasing later analysis. Second, it ensures that the final dataset used for domain classification is less noisy and more representative of unique grievance instances.

Deduplication is especially useful in public grievance datasets because citizens may submit similar or repeated complaints through multiple channels or at different points in time. Removing exact duplicates helps the downstream models focus on genuinely distinct complaint narratives.

4.9 Preparation of the Final Cleaned Dataset

The output of the preprocessing notebook is a consolidated grievance dataset with normalized text, cleaned metadata, and reduced duplication. This cleaned dataset becomes the working corpus for the later domain-classification stage.

At this point in the workflow, the dataset has the following properties:

- structurally consistent columns;
- standardized datatypes for dates and text;
- reduced missingness in non-critical fields;
- normalized grievance text suitable for machine learning;
- deduplicated complaint records for cleaner analysis.

Chapter 5

Proposed Methodology

5.1 Introduction

The proposed methodology follows a sequential pipeline architecture in which each stage performs one well-defined analytical function. Instead of building a monolithic model, the project decomposes the grievance-processing problem into multiple subproblems and solves each one separately. This design is suitable for grievance redressal because the real-world process itself is multi-stage: a complaint must first be sanitized, then examined for duplication, then categorized, then summarized, then prioritized, and finally assigned an expected service duration.

5.2 End-to-End Processing Logic

Algorithm 1: End-to-End Grievance Processing Pipeline

Input: Raw grievance dataset
Output: Final structured grievance output
Load raw data
Apply preprocessing and text normalization
Perform anomaly detection to remove duplicates or near-duplicates
Classify the grievance into an organizational domain
Extract keywords and generate short summaries
Predict severity / urgency category
Predict expected resolution time
Write final output to structured CSV file

5.3 Anomaly Detection Module

5.3.1 Objective

The objective of the anomaly detection module is to ensure that only meaningful grievance records are processed by the remaining stages of the pipeline. This module performs two important functions: content validation and duplicate detection. The first step removes spam, abusive, irrelevant, or very short text inputs, while the second step identifies duplicate or highly similar grievances.

5.3.2 Content Validation and Filtering

Before similarity-based anomaly detection is applied, the system validates the input text using a zero-shot classification model. Each grievance is assigned scores for multiple semantic categories, including:

- valid public service complaint or grievance,
- spam, advertisement, or scam message,
- abusive, offensive, or threatening language,
- irrelevant or meaningless text.

The input text is first stripped of extra whitespace and checked for length. Very short texts, especially those with two or fewer words, are rejected immediately because they usually do not contain sufficient information for analysis.

For the remaining records, the classifier produces confidence scores for the candidate labels. The system applies threshold-based rules to reject inputs that are likely to be spam, abusive, or irrelevant. Specifically, a grievance is rejected if:

- the spam score is greater than or equal to 0.80 and also greater than the valid grievance score;
- the irrelevant score is greater than or equal to 0.75 and also greater than the valid grievance score;
- the abusive score is greater than or equal to 0.85 and also greater than the valid grievance score.

Only grievances that satisfy the acceptance criteria are forwarded to the duplicate detection stage. This filtering step improves data quality and prevents noisy or malicious content from affecting downstream modules.

5.3.3 Blocking Strategy

After validation, the duplicate detection stage applies a blocking strategy to reduce the search space for similarity comparison. Instead of comparing every new grievance against the complete historical database, the system considers only a smaller subset of records based on temporal and administrative conditions.

In this implementation, the comparison is restricted to grievances that satisfy the following constraints:

- the grievance must fall within a 30-day time window ending at the received date of the new record;
- the grievance must belong to the same `state`;
- the grievance must share the same `_id`;
- the incoming record must be accepted by the content validation layer.

This blocking strategy reduces the number of pairwise comparisons while preserving relevant contextual similarity. It also ensures that the anomaly detection step remains efficient when applied to a large grievance history.

5.3.4 TF-IDF Similarity Calculation

After blocking, the grievance text is transformed into TF-IDF feature vectors. Cosine similarity is then computed between the new grievance and each historical candidate. This allows the module to measure textual closeness based on the importance of shared words and phrases.

The similarity scores are interpreted in two ways depending on the execution stage. In the exploratory training notebook, the top matching complaints are ranked and categorized as follows:

- **Exact / High Duplicate** for similarity ≥ 0.85;
- **Potential Duplicate** for similarity between 0.60 and 0.85;
- **Unique** for similarity < 0.60.

In the pipeline notebook, the maximum similarity score is used as the final decision criterion. If the maximum similarity score exceeds 0.95, the grievance is treated as a duplicate. Otherwise, it is considered unique and written to the next-stage input file.

5.3.5 Role in the Pipeline

The anomaly detection module acts as a quality-control layer in the grievance processing pipeline. By filtering out duplicate, highly similar, spam, abusive, or irrelevant complaints, it ensures that the downstream modules operate on cleaner and more meaningful data. This improves the reliability of complaint classification, keyword extraction, severity scoring, and resolution-time prediction.

5.4 Domain Classification Module

5.4.1 Objective

The domain classification module is responsible for assigning each grievance to the most relevant organizational domain or department. This step is critical in the grievance redressal pipeline, as accurate classification ensures that complaints are routed to the appropriate authority for timely resolution. Incorrect classification may lead to delays, misrouting, and inefficiencies in the redressal process.

5.4.2 Feature Representation using TF-IDF

Before applying machine learning models, the grievance text is transformed into a numerical representation using **Term Frequency–Inverse Document Frequency (TF-IDF)**. TF-IDF captures the importance of words in a document relative to the entire corpus.

The TF-IDF representation is computed as:

- **Term Frequency (TF):** Measures how frequently a term appears in a document.
- **Inverse Document Frequency (IDF):** Reduces the weight of commonly occurring words and increases the importance of rare but informative words.

In this project, **n-gram features** (unigrams and bigrams) are used to capture both individual words and short phrases. This is particularly useful for grievance text, where phrases such as “water supply” or “electricity issue” carry more meaning than isolated words.

5.4.3 Training Strategy

The training notebook evaluates two classical linear models for multi-class text classification:

- **Logistic Regression**
- **Linear Support Vector Machine (SVM)**

Both models are trained on TF-IDF feature vectors with **class-balanced weighting** to address class imbalance in the dataset. This ensures that minority classes are not ignored during training.

The trained model is serialized and stored as `svm_domain_classifier.pkl`, enabling efficient reuse during inference.

5.4.4 Inference Strategy

During inference, the saved classifier is loaded and applied directly to the processed grievance text. The predicted label is written into the `org_code` field so that downstream modules can use a consistent domain prediction.

5.4.5 Methodological Contribution

The domain classifier is intentionally lightweight and classical rather than overly complex. This is appropriate because grievance text is often sparse, domain-specific, and more effectively handled by linear decision boundaries over TF-IDF features than by highly elaborate architectures.

5.4.6 Role in the Pipeline

The domain classification module serves as the foundation for the entire grievance processing pipeline. By assigning each complaint to the correct domain, it enables:

- automatic routing of grievances to relevant departments,
- domain-aware keyword extraction and analysis,
- more accurate severity and time prediction based on contextual information.

Thus, this module plays a central role in transforming unstructured complaint text into actionable administrative insights.

5.5 Keyword Extraction and Summarization Module

5.5.1 Objective

The keyword extraction module is designed to identify the most representative terms and phrases present in a grievance. Since grievance text is often lengthy and unstructured, extracting key information helps in quickly understanding the core issue. The output of this module improves interpretability, supports summarization, and assists in reporting and downstream analysis.

5.5.2 YAKE-Based Extraction

The system employs **YAKE (Yet Another Keyword Extractor)** for unsupervised keyword extraction. YAKE is a statistical method that does not require any external corpus or prior training. Instead, it relies on features derived directly from the input text.

YAKE evaluates candidate keywords based on several factors, including:

- term frequency within the document,
- position of the term (early occurrences are often more important),
- word casing and formatting,
- context diversity and co-occurrence patterns.

Lower YAKE scores indicate more important keywords. In this implementation, the algorithm extracts short phrases of up to two tokens and returns the top-ranked keywords. YAKE is particularly suitable for grievance data because it adapts well to domain-specific language without requiring pre-training.

5.5.3 TextRank-Based Extraction

In addition to YAKE, the system uses **TextRank**, a graph-based ranking algorithm inspired by the PageRank algorithm. TextRank constructs a graph where:

- nodes represent words or phrases,
- edges represent co-occurrence relationships within a fixed window of text.

Each node is assigned an importance score based on its connections with other nodes. Words or phrases that frequently co-occur with other important words receive higher scores.

The implementation uses the **spaCy TextRank pipeline**, which efficiently extracts key phrases from the processed grievance text. Unlike YAKE, which relies on statistical features, TextRank captures the structural relationships between words, providing a complementary perspective on keyword importance.

5.5.4 LexRank Summarization

For summarization, the system applies **LexRank**, another graph-based algorithm designed for extractive summarization. LexRank operates at the sentence level rather than the word level.

The process involves:

- representing each sentence as a TF-IDF vector,
- computing cosine similarity between sentence pairs,
- constructing a similarity graph where sentences are nodes,
- ranking sentences based on centrality within the graph.

Sentences that are highly connected to other sentences are considered more informative. The top-ranked sentences are selected to form the summary.

In this implementation, if a grievance contains sufficient length, LexRank generates a concise two-sentence summary. This is particularly useful for long complaints where manual reading would be time-consuming.

5.5.5 Role in the Pipeline

The keyword extraction and summarization module serves two important purposes within the overall pipeline. First, it enhances interpretability by allowing administrators to quickly grasp the essence of a complaint without reading the entire text. Second, it provides a compact semantic representation of the grievance, which can support downstream tasks such as severity scoring and reporting.

By combining YAKE, TextRank, and LexRank, the system leverages both statistical and graph-based techniques, resulting in a more robust and comprehensive understanding of grievance content.

5.6 Severity Risk Scoring Module

5.6.1 Objective

The severity risk scoring module is designed to estimate the urgency level of each grievance. This is a critical component of the system, as it enables prioritization of complaints based on their importance, ensuring that critical issues are addressed promptly.

5.6.2 Feature Construction

The severity model uses a combination of textual and contextual features to capture both the semantic content of the complaint and its administrative context. The features include:

- processed grievance text;
- organization code (`org_code`);
- demographic attribute (`sex`);
- geographic attribute (`state`);
- temporal features such as month of receipt;
- day of the week of receipt.

The grievance text is vectorized using TF-IDF, while categorical metadata is encoded using one-hot encoding. This hybrid feature representation allows the model to learn richer patterns compared to using text alone.

5.6.3 Model Choice

The training process evaluates multiple classification models, including Logistic Regression, Linear SVM, Random Forest, and XGBoost. Among these, the **Linear Support Vector Machine (SVM)** demonstrates the best performance.

SVM is particularly well-suited for this task because:

- it performs effectively in high-dimensional sparse feature spaces;
- it provides strong generalization for text classification tasks;
- it handles class imbalance effectively when combined with class weighting.

The final model is implemented as a pipeline combining TF-IDF vectorization, feature encoding, and classification, and is serialized as `urgency_model_svc.pkl` for use during inference.

5.6.4 Reason for Metadata Inclusion

Incorporating contextual metadata enhances the predictive capability of the model beyond text-only features. Grievance urgency is often influenced by factors such as:

- temporal patterns (e.g., delays during specific periods);
- department-specific processing behavior;
- regional administrative differences.

By integrating calendar and demographic attributes with textual data, the model gains a more comprehensive understanding of grievance patterns. This improves its ability to accurately distinguish between high, medium, and low urgency complaints.

5.7 Resolution Time Prediction Module

5.7.1 Objective

The resolution time prediction module estimates the number of days required to resolve a grievance. This is an important part of the grievance redressal pipeline because it helps administrators understand workload, manage pending complaints, and set realistic expectations regarding service completion.

5.7.2 Target Construction and Data Cleaning

The target variable, `resolution_days`, is computed as the difference between `closing_date` and `recvd_date`. Since the dataset may contain invalid records due to data-entry issues, complaints with negative resolution days are removed. In addition, outlier records are filtered using the Interquartile Range (IQR) method so that extremely large resolution times do not distort the regression model.

After outlier removal, the dataset size is reduced from 37,854 rows to 34,608 rows, which results in a cleaner and more reliable training corpus.

5.7.3 Feature Construction

The final model uses a combination of textual, categorical, and temporal features. The selected input features are:

- `grievance_text_processed`;
- `org_code`;
- `sex`;
- `state`;
- `recvd_month`.

The grievance text is transformed using TF-IDF vectorization, while categorical features are encoded separately. The month of complaint receipt is included as a temporal feature because resolution time may vary with seasonal or administrative patterns.

5.7.4 Model Selection Strategy

Multiple regression models were evaluated for this task, including Ridge Regression, Random Forest Regressor, and XGBoost Regressor. The notebook also explores GPU-based hyperparameter tuning using `RandomizedSearchCV` over parameters such as `n_estimators`, `learning_rate`, `max_depth`, `subsample`, and `colsample_bytree`.

Among the tested models, **XGBoost** produces the best practical performance on the cleaned dataset. The final trained pipeline is serialized as `time_predict.pkl` for reuse during inference.

5.7.5 Reason for Feature Choice

The inclusion of both grievance text and contextual metadata improves prediction quality because resolution time is influenced not only by what the complaint says, but also by the administrative domain, complainant context, and the time of receipt. This hybrid representation enables the model to capture patterns that are not visible from text alone.

5.8 Sentiment Analysis on Feedback Data

5.8.1 Objective

The sentiment analysis module is designed to classify grievance feedback into sentiment categories that reflect the citizen's overall experience after or during complaint handling. In the context of a grievance redressal system, sentiment provides an additional interpretive layer beyond domain classification and severity scoring.

It helps capture whether the citizen feels satisfied, neutral, or dissatisfied with the response or outcome of the complaint.

5.8.2 Label Construction through Weak Supervision

Since the available feedback data is unlabeled, the module begins with a weak supervision stage to generate initial sentiment labels. A rule-based keyword strategy is used to assign one of three sentiment classes: `satisfied`, `neutral`, and `unsatisfied`. The labeling process relies on domain-specific positive and negative expressions such as *resolved*, *fixed*, *helpful*, *not resolved*, *delay*, *pending*, and *no response*.

5.8.3 Text Preprocessing

Before model training, the feedback text is normalized using standard preprocessing operations. These include lowercasing, removal of URLs, punctuation, and non-alphabetic characters, as well as whitespace normalization. The preprocessing step reduces noise and ensures that the model focuses on meaningful lexical patterns in the feedback.

5.8.4 Feature Construction

The cleaned feedback text is transformed into numerical features using TF-IDF vectorization. A unigram-bigram representation is used so that both isolated words and short phrases can be captured. This is useful in sentiment classification because expressions such as *no response*, *very satisfied*, and *not resolved* carry stronger semantic signals than single words alone.

5.8.5 Model Selection Strategy

Multiple linear text classifiers are evaluated for this task, including Logistic Regression, Linear Support Vector Machine (SVM), SGD Classifier, Multinomial Naive Bayes, and Complement Naive Bayes. The selection is based on performance on a held-out test split using accuracy and macro-averaged F1-score as the key metrics.

Among the tested models, Linear SVM produces the best overall performance. This is consistent with the common observation that linear classifiers perform well on sparse TF-IDF representations, especially for text classification tasks with moderately balanced classes.

5.8.6 Role in the Pipeline

The sentiment module is intended to operate on feedback data as a separate but complementary stage within the grievance redressal pipeline. While domain classification identifies the department and severity scoring estimates urgency, sentiment analysis captures the user's emotional response to the redressal process. Together, these modules provide a more complete understanding of the grievance lifecycle and improve the system's ability to support administrative decision-making.

5.9 Mathematical and Operational Interpretation

The methodology can be understood as a transformation chain:

$$\text{Raw Complaint} \rightarrow \text{Clean Text} \rightarrow \text{Unique Record} \rightarrow \text{Domain} \rightarrow$$

$$\text{Keywords} \rightarrow \text{Urgency} \rightarrow \text{Resolution Time}$$

Each step reduces uncertainty and increases structure. The final output is therefore not merely a textual complaint, but a data-rich administrative object that can be routed and prioritized.

Chapter 6

Implementation Details

6.1 Introduction

The implementation is organized as a set of Jupyter notebooks, each notebook corresponding to one pipeline stage. This design is beneficial because it allows each module to be developed, executed, and validated independently before being connected to the rest of the system. The project follows a modular notebook-based workflow for the main grievance processing pipeline, while the sentiment analysis component is maintained as a separate but integrated module for feedback analysis.

6.2 Project Directory and Module Mapping

Table 6.1: Mapping between project folders and functional modules

Folder / File	**Function**
`00-Data-Acquisition-Analysis`	Exploratory analysis and understanding data
`01-Data-Preprocessing`	Text cleaning, normalization, and dataset filtering
`02-Anomaly-Detection`	Duplicate and near-duplicate grievance filtering
`03-Domain-Classification`	Complaint domain prediction
`04-Keyword-Extraction`	Keyword extraction and extractive summarization
`05-Severity-Risk-Scoring`	Urgency prediction and severity classification
`06-Resolution-Time-Prediction`	Resolution-time regression and SLA classification
`07-Sentiment-Analysis`	Feedback sentiment analysis module
`run_pipeline.py`	Sequential execution of the pipeline notebooks

Repository Link: Automated-Grievance-Redressal-System-Project

6.3 Pipeline Execution Mechanism

The file `run_pipeline.py` defines a list of notebooks and executes them sequentially using `jupyter nbconvert`. This means that each notebook is run in order, and intermediate files are written to disk after every stage. The implementation is simple, reproducible, and directly aligned with the conceptual architecture of the project.

The sentiment analysis module is implemented through separate notebooks, namely `Sentiment_Analysis_training.ipynb` and `Sentiment_Analysis_pipeline.ipynb`. Since this module operates on feedback data rather than the main grievance-processing input, it is maintained as an auxiliary analytical track while still remaining fully compatible with the overall project structure.

6.4 Model Serialization and Reuse

The trained classifiers and regressors are saved in `.pkl` format. This provides two practical benefits. First, expensive training need not be repeated every time the system is run. Second, the inference notebooks can load a compact model artifact and apply it directly to new grievance data. The following models used:

- `svm_domain_classifier.pkl`
- `urgency_model_svc.pkl`
- `time_predict.pkl`
- `sentiment_predict.pkl`

6.5 Implementation of Sequential Data Handoffs

Each module writes its output to an intermediate CSV file. This creates a transparent chain of dependencies:

- preprocessing output → anomaly detection input;
- anomaly detection output → domain classification input;
- domain classification output → keyword extraction input;
- keyword extraction output → severity scoring input;
- severity scoring output → resolution-time prediction input;
- final output → `07_output.csv`.

The sentiment analysis module follows a parallel workflow on feedback data:

- raw feedback input → text cleaning and weak labeling;
- labeled feedback → TF-IDF feature extraction and model training;
- trained sentiment model → prediction on feedback text;
- final sentiment output → `final_sentiment_output.csv`.

6.6 Sentiment Module Integration

The sentiment analysis component extends the grievance redressal system by analyzing citizen feedback associated with complaint resolution. Unlike the main grievance pipeline, which focuses on domain classification, severity scoring, and resolution-time estimation, the sentiment module captures the user's perception of the service outcome.

The sentiment workflow uses weak supervision to assign initial labels such as `satisfied`, `neutral`, and `unsatisfied`. These labels are then used to train a supervised text classifier on the cleaned feedback corpus. Among the evaluated classifiers, Linear SVM is selected as the final model due to its strong performance on sparse TF-IDF representations.

This module is implemented separately so that it can be trained, tested, and updated independently from the main grievance-processing pipeline. Such separation improves maintainability and allows sentiment analysis to evolve further with manual annotation or more advanced language models in future iterations.

6.7 Reproducibility Considerations

The notebook-based implementation makes the project easy to reproduce. Every stage can be rerun independently, and the saved model artifacts allow the final pipeline to be executed without retraining. This is especially useful where demonstration, experimentation, and iterative improvement are all important.

The separation of the sentiment module into its own notebooks also improves reproducibility, because feedback analysis can be updated independently without affecting the main grievance-processing workflow.

Chapter 7

Results and Discussion

7.1 Experimental Setup

The project was implemented using Python and a combination of classical machine learning and text-mining libraries. The main experimental workflow involved training models on grievance data, validating them using standard train-test splits, and then applying the saved models to the pipeline output. For classification tasks, metrics such as accuracy, precision, recall, and macro F1-score were used. For regression tasks, mean absolute error, root mean squared error, and coefficient of determination were used.

7.2 Preprocessing and Filtering Results

The preprocessing stage successfully filtered the raw dataset using content validation, language constraints, minimum length requirements, and status-based exclusion. The resulting dataset is more consistent and suitable for downstream natural language processing tasks. The text normalization steps further improve token consistency by removing case variation, markup artifacts, URLs, punctuation, and common stopwords.

7.3 Anomaly Detection Results

The anomaly detection module implements a two-stage filtering strategy based on content validation and TF-IDF cosine similarity. For each new grievance, the system first checks whether the text is meaningful enough to process. Very short inputs are rejected immediately, and the zero-shot classifier is used to filter out spam, abusive, and irrelevant content. Only valid grievances are forwarded to the duplicate detection stage.

For duplicate detection, the module considers only historical records from the same `state` and `_id` within a 30-day time window. This reduces the search space and makes the comparison process computationally efficient. Cosine similarity is then computed between the new grievance and the candidate historical records. In the exploratory notebook, the resulting similarity scores are interpreted as *Exact / High Duplicate*, *Potential Duplicate*, or *Unique*. In the pipeline execution, a maximum similarity score greater than 0.95 is used to mark a record as duplicate.

This approach effectively prevents redundant and low-quality grievances from propagating through the pipeline. It also provides a more interpretable output than a simple binary decision, since the administrator can distinguish between strong duplication, partial overlap, and invalid input. Overall, the anomaly detection module improves both the quality and reliability of the grievance processing system.

7.4 Domain Classification Results

The domain classification module was evaluated using two linear text classifiers trained on TF-IDF features: Logistic Regression and Linear SVM. Both models used the same train-test split with class balancing, and the task was to predict the grievance domain represented by `org_code`. The results show that the Linear SVM model performs better than Logistic Regression on this multi-class text classification task.

Table 7.1: Domain classification performance comparison

Model	**Accuracy**	**Macro F1-score**
Logistic Regression	0.9666	0.8488
Linear SVM	0.9900	0.9078

The class-wise report indicates that the model performs extremely well on the major grievance domains with large support values, while a few very small classes show unstable precision or recall due to limited training samples. This behavior is expected in highly imbalanced multi-class text classification problems. Overall, the results confirm that TF-IDF features combined with a linear SVM classifier are highly effective for grievance domain prediction and automated routing.

The final trained classifier was serialized and saved as `svm_domain_classifier.pkl` for use in the inference pipeline.

Table 7.2: Representative class-wise performance of the final Linear SVM model

Class	**Precision**	**Recall**	**F1-score**
CBODT	1.00	1.00	1.00
CBOEC	0.98	1.00	0.99
DARPG	1.00	0.98	0.99
DEABD	1.00	0.99	1.00
DEAID	0.99	0.99	0.99
DOCAF	0.99	1.00	1.00
DCOYA	0.96	1.00	0.98
DHLTH	0.99	0.96	0.98

7.5 Severity Risk Scoring Results

The severity risk scoring module was evaluated using multiple machine learning models trained on a combined feature space consisting of processed grievance text and contextual metadata. The models compared include Logistic Regression, Linear Support Vector Machine (SVM), Random Forest, and XGBoost.

The results indicate that the Linear SVM model performs the best among all evaluated models due to its effectiveness in handling high-dimensional sparse text features.

Table 7.3: Severity classification performance comparison

Model	**Accuracy**	**Macro F1-score**
Logistic Regression	0.90	0.87
Linear SVM	0.94	0.93
Random Forest	0.92	0.90
XGBoost	0.50	0.36

The Linear SVM classifier achieved an accuracy of approximately 94% and a macro F1-score of approximately 93%, indicating strong performance across all urgency classes.

The results demonstrate that urgency can be reliably inferred using a combination of textual and contextual features. The model performs particularly well on the dominant `medium` class while maintaining strong performance for `high` and `low` urgency categories. This enables effective prioritization of grievances in real-world scenarios.

Table 7.4: Class-wise performance of the final severity model

Class	Precision	Recall	F1-score	Support
high	0.97	0.90	0.93	1551
low	0.85	0.95	0.90	1181
medium	0.96	0.96	0.96	4839
Accuracy		0.94		
Macro Avg. F1		0.93		
Weighted Avg. F1		0.94		

7.6 Resolution Time Prediction Results

The resolution time prediction module was evaluated using multiple regression models on the cleaned grievance dataset. The target variable was derived from the difference between the grievance closing date and the received date, and the final evaluation was performed after removing negative and outlier resolution times.

The following table summarizes the performance of the tested models:

Table 7.5: Resolution time prediction performance comparison

Model	MAE (days)	RMSE (days)	R^2
Ridge Regression	15.55	24.91	0.3490
Random Forest Regressor	13.31	23.70	0.4106
XGBoost (earlier experiment)	14.27	24.24	0.3834
XGBoost (final cleaned-data model)	7.61	10.53	0.4094

The final XGBoost model trained on the cleaned dataset achieves the best practical performance, with a mean absolute error of approximately 7.61 days and an RMSE of approximately 10.53 days. This shows that the model is able to estimate grievance resolution time with useful accuracy for administrative planning.

The reduction in error after outlier removal indicates that noisy and extreme resolution values were negatively affecting the earlier models. By cleaning the target distribution and retaining only valid records, the model was able to learn more stable patterns from the data.

Although the R^2 score remains moderate, the low MAE and RMSE values show that the system is sufficiently useful for practical estimation of grievance resolution duration. In operational settings, this can help administrators identify potentially long-pending grievances and manage service-level expectations more effectively.

The final trained pipeline was saved as `time_predict.pkl` for use in the inference stage.

7.7 Keyword Extraction and Interpretability

The keyword extraction module does not produce a single numeric score, but it adds a major interpretability benefit. By extracting YAKE keywords, TextRank phrases, and LexRank summaries, the system makes each grievance more readable to a human reviewer. This is useful in government or administrative workflows, where reviewers often need to understand the complaint quickly without reading the full record.

7.8 Sentiment Analysis Results

The sentiment classification module was evaluated using multiple linear and probabilistic classifiers on the weakly labeled feedback dataset. The final comparison shows that the Linear SVM model performs the best among all tested algorithms.

Table 7.6: Sentiment classification performance comparison

Model	**Accuracy**	**Macro F1-score**
Linear SVM	0.8363	0.8379
SGD Classifier	0.8279	0.8298
Logistic Regression	0.8238	0.8249
Complement Naive Bayes	0.6609	0.6620
Multinomial Naive Bayes	0.6498	0.6093

The Linear SVM classifier achieves an accuracy of approximately 83.63% and a macro F1-score of approximately 0.8379, making it the best-performing model for sentiment prediction in this setup. The performance is balanced across the three sentiment categories and is stronger than the probabilistic baselines, which are less effective for the sparse TF-IDF feature space.

The moderate reduction in accuracy compared to earlier experiments is expected because the final dataset is generated using weak supervision and contains real-world noise. However, the macro F1-score remains strong, indicating that the model does not rely excessively on a single dominant class and instead learns useful distinctions across satisfied, neutral, and unsatisfied feedback.

The final sentiment model was serialized for reuse in the pipeline into the grievance redressal pipeline. It can be used to analyze citizen feedback and provide an additional indicator of service quality and complaint resolution effectiveness.

7.9 Discussion

The overall results indicate that the project successfully converts unstructured grievance and feedback data into structured, decision-support-oriented outputs. Several observations can be made:

- Data preprocessing significantly improves input quality and reduces textual noise.
- Duplicate filtering prevents repeated grievances from contaminating the analytical pipeline.
- Linear SVM is highly effective for text-based severity, domain classification, and sentiment tasks.
- XGBoost is a strong choice for resolution-time prediction on mixed feature spaces.
- Keyword extraction improves transparency and human interpretability.
- Sentiment analysis adds an important satisfaction-oriented perspective to grievance feedback.

The results also suggest an important practical insight: grievance redressal is best approached as a multi-stage analytics problem rather than a single classification task. The combination of routing, prioritization, keyword extraction, sentiment analysis, and resolution estimation gives the system administrative relevance that a simple text classifier would not provide. By integrating sentiment analysis on feedback data, the pipeline becomes more complete and capable of capturing not only the administrative status of a grievance but also the user's perception of the resolution process.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This project proposed and implemented a modular grievance redressal analytics pipeline designed to transform raw, unstructured complaint text into structured and actionable information for administrative use. The system was developed as a sequence of well-defined stages, beginning with data acquisition and preprocessing and proceeding through anomaly detection, domain classification, keyword extraction, severity risk scoring, and resolution-time prediction. This staged design is aligned with the practical workflow of grievance-handling systems, where complaints must first be cleaned, then examined for redundancy, categorized into the appropriate domain, summarized through keywords, prioritized according to urgency, and finally associated with an estimated resolution horizon [1, 2, 3, 4, 5, 6].

The implemented system demonstrates that grievance redressal can be treated as a structured natural language processing problem rather than only as a record-management problem. By using classical machine learning techniques such as TF-IDF-based text representation, Support Vector Machines, and XGBoost, the project achieves a balance between interpretability, practical performance, and implementation simplicity. The project therefore shows how a grievance platform can be made more intelligent without introducing unnecessary complexity [9, 7, 8].

One of the important outcomes of the work is that the system does not rely on a single classifier or predictor. Instead, it uses a pipeline architecture in which each module contributes a specific function to the overall decision-support process. This design is significant because it mirrors the real-world grievance lifecycle more closely than a monolithic model would. The preprocessing module improves text quality, the anomaly detection module prevents duplicate records from propagating, the domain classifier supports routing, the keyword extraction module improves interpretability, the severity module enables prioritization, and the time prediction module supports planning and monitoring. As a result, the final output is not merely a classification label but a structured representation of a grievance that is more useful for operational decision-making [10, 3, 4].

Overall, the project achieves its main objective of building a formal, modular, and technically sound grievance analytics pipeline. It demonstrates that natural language processing and machine learning can be combined effectively to improve classification, prioritization, summarization, and time estimation in complaint-processing environments. In doing so, the system contributes a practical academic model for intelligent grievance management and establishes a clear base for future enhancements.

8.2 Key Contributions of the Project

The main contributions of this work can be summarized as follows:

- a modular end-to-end grievance processing pipeline was designed and implemented;
- complaint data was cleaned, normalized, and filtered for downstream NLP tasks;
- a blocking-based anomaly detection strategy was introduced to reduce duplicate propagation;
- a domain classification model was built for automated grievance routing;
- a keyword extraction and summarization layer was added for interpretability;
- a severity risk scoring module was developed to support complaint prioritization;
- a resolution-time prediction module was created for administrative planning;
- a sentiment analysis module was integrated for feedback data.

8.3 Future Scope

The project offers several promising directions for further enhancement. Since the present system is built as a modular pipeline, new analytical components can be added without redesigning the entire architecture.

A first major extension is multimodal grievance analysis. Recent research has shown that complaint systems become more powerful when they can combine text with images, especially in cases where the complaint is supported by visual evidence such as damaged infrastructure, sanitation problems, or service-related defects [2]. Adding image processing would make the system more suitable for modern grievance portals where citizens may upload both text and photographs.

A second direction is multilingual support. Many public grievance systems must handle complaints in multiple languages, dialects, or mixed-language forms. Extending the current pipeline to support multilingual text preprocessing and classification would significantly improve accessibility and usability.

A third future enhancement is the use of transformer-based language models and large language models for more context-aware understanding of complaint text. While the current system uses classical machine learning methods for interpretability and efficiency, transformer-based methods could improve performance on longer, more complex, or more ambiguous grievances. Such models may be especially useful for advanced summarization, semantic matching, and conversational complaint assistance [6, 10].

A fourth direction is deployment in a real-time web application or dashboard. The current project already provides a strong pipeline foundation, and exposing it through a user-facing interface would make the system more practical for administrators. A dashboard could display complaint counts, severity distribution, domain-wise trends, and predicted resolution times, thereby enabling operational monitoring and decision support.

Finally, future work may include integration with automated escalation policies, service-level agreement monitoring, and department-wise analytics. These additions would transform the system from a complaint-processing prototype into a more complete grievance intelligence platform capable of supporting public administration at scale.

8.4 Final Remark

In conclusion, the project demonstrates that grievance redressal can be significantly improved through the application of structured natural language processing and machine learning techniques. By separating the problem into modular stages and preserving future extensibility, the work provides a practical implementation. The current system already addresses several core needs of grievance processing, and its architecture makes it well suited for future development.

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